

ENGINEERING CALCULATION SHEET

Minimum Required Wall Thickness & MAWP Check - Process Piping

Calc No.:	CALC-U300-PL-014	Rev:	2
Line No.:	8"-P-1043-A1A	Unit:	300
Service:	Hydrocarbon Liquid (Naphtha)	Design Code:	ASME B31.3
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1. Objective

Determine the minimum required wall thickness for line 8"-P-1043-A1A (surge drum V-301 outlet to stripper column T-501) per ASME B31.3 straight pipe under internal pressure, and verify the selected schedule against the design pressure with corrosion allowance.

2. Design Basis / Input Data

Parameter	Symbol	Value	Unit
Design Pressure	P	10.5	kg/cm2(g)
Design Temperature	T	180	deg C
Pipe Outside Diameter (8" Sch)	D	219.1	mm
Allowable Stress at Design Temp (A106 Gr.B)	S	1005	kg/cm2
Weld Joint Efficiency (seamless)	E	1.00	-
Y-coefficient (ferritic steel, T < 480C)	Y	0.40	-
Corrosion Allowance	C	1.5	mm
Mill Tolerance	-	12.5	%

3. Governing Equation (ASME B31.3, Eq. 3a)

Pressure design thickness for $t < D/6$:

$$t = (P \times D) / (2 \times (S \times E + P \times Y))$$

Minimum required thickness including corrosion allowance:

$$t_{\min} = t + C$$

4. Calculation Steps

Step 1 - Convert design pressure to consistent units:

$$P = 10.5 \text{ kg/cm}^2 = 10.30 \text{ kgf/mm}^2 \approx 1.030 \text{ N/mm}^2 \text{ (MPa)}$$

Step 2 - Convert allowable stress to consistent units:

$$S = 1005 \text{ kg/cm}^2 = 98.55 \text{ N/mm}^2 \text{ (MPa)}$$

Step 3 - Apply the governing equation with $D = 219.1 \text{ mm}$, $E = 1.00$, $Y = 0.40$:

$$t = (1.030 \times 219.1) / (2 \times (98.55 \times 1.00 + 1.030 \times 0.40))$$

$$t = 225.68 / (2 \times (98.55 + 0.412))$$

$$t = 225.68 / 197.92$$

$$t = 1.140 \text{ mm}$$

Step 4 - Add corrosion allowance:

$$t_{\min} = t + C = 1.140 + 1.5 = 2.640 \text{ mm}$$

Step 5 - Compare against nominal wall thickness of selected pipe.

Selected: 8" NB, Schedule 40, nominal wall $t_{\text{nom}} = 8.18 \text{ mm}$

Mill tolerance (12.5%) minimum supplied wall:

$$t_{\text{supplied_min}} = t_{\text{nom}} \times (1 - 0.125) = 8.18 \times 0.875 = 7.158 \text{ mm}$$

Step 6 - Margin check:

Margin = $t_{\text{supplied_min}} - t_{\min} = 7.158 - 2.640 = 4.518 \text{ mm}$ (PASS, large margin)

5. MAWP Check (Reverse Calculation)

Using the supplied minimum wall thickness (post mill-tolerance, before corrosion), back-calculate the Maximum Allowable Working Pressure to confirm it exceeds the design pressure:

$$P_{\text{MAWP}} = (2 \times S \times E \times t) / (D - 2 \times Y \times t)$$

$$t_{\text{effective}} = t_{\text{supplied_min}} - C = 7.158 - 1.5 = 5.658 \text{ mm}$$

$$P_{\text{MAWP}} = (2 \times 98.55 \times 1.00 \times 5.658) / (219.1 - 2 \times 0.40 \times 5.658)$$

$$P_{\text{MAWP}} = 1115.3 / (219.1 - 4.526)$$

$$P_{\text{MAWP}} = 1115.3 / 214.57$$

$$P_{\text{MAWP}} = 5.198 \text{ N/mm}^2 = 53.0 \text{ kg/cm}^2(\text{g})$$

6. Result Summary

Check	Required	Available	Result
Min. wall thickness (incl. CA)	2.640 mm	7.158 mm	PASS
MAWP at selected schedule	10.5 kg/cm2(g)	53.0 kg/cm2(g)	PASS

7. Conclusion

8-inch Schedule 40 seamless pipe (ASTM A106 Gr.B) is adequate for line 8"-P-1043-A1A at the stated design conditions, with a comfortable margin on both minimum wall thickness and MAWP. No upgrade in schedule is required. This calculation should be re-verified if design temperature, service, or corrosion allowance assumptions change.

Note: Allowable stress values per ASME B31.3 Table A-1 for A106 Gr.B at 180 deg C (interpolated). This is a worked training/test example - values are representative, not for construction use.